

Binary Search

Week 2, Lecture 1

A demo lecture for transcript-per-slide

Searching a sorted list

- Linear search: up to n checks
- Sorted order lets us skip most of them

The idea

- Look at the middle element
- Too small? Discard the left half
- Too big? Discard the right half
- Repeat on what is left

Example: find 23

[2 5 8 12 16 23 38 56 72 91]

mid = 16 -> go right

mid = 56 -> go left

mid = 23 -> found

Why about $\log_2 n$ steps

$n \rightarrow n/2 \rightarrow n/4 \rightarrow \dots \rightarrow 1$
1,000,000 items: about 20 steps

Off-by-one traps

```
while lo <= hi:  
    mid = lo + (hi - lo) // 2
```

Assignment 1

- Due Friday week 4, 5pm
- Worth 15%
- Submit on the course website

Summary

- Sorted input + halving = $O(\log n)$
- Get the loop bounds right